

Five observations on five years of contact hypothesis research

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Abstract

I offer five observations on the state of contact hypothesis research five years after the publication of [The Contact Hypothesis Re-evaluated](#). First, rigorous field experiments have proliferated, and they often find conflicting, disappointing results. Second, to explain those conflicting results, I propose a theory of multiple forms of prejudice, some of which are amenable to contact and some of which are not. Third, light touch interventions are appropriate for light prejudices. Fourth, assimilation is an undertheorized moderator of contact. Fifth, we would still benefit from more systematic tests of Gordon Allport's moderating conditions: shared goals, cooperation, equal status, and institutional support.

Introduction: Seven decades of the contact hypothesis

Gordon Allport's [The Nature of Prejudice](#) (1954), is best remembered today for proposing the contact hypothesis, the idea that contact between groups can reduce tensions between them. However, Allport did not believe that just "casual contact" would do. In order to create "true acquaintance," he argued, contact should meet four additional conditions: shared goals, cooperation, equal status, and institutional support, "i.e., by law, custom or local atmosphere" (p. 281). Otherwise, he thought, "the more contact the more trouble" (p. 263).

Nearly two decades ago, an influential meta-analysis by [Pettigrew and Tropp \(2006\)](#) reviewed hundreds of studies and found that contact "typically reduces intergroup prejudice" (p. 751). Moreover, Allport's conditions were helpful but unnecessary: "samples with no claim to these key conditions still show significant relationships between contact and prejudice" (p. 766). Overall, they write, "[t]here is little need to demonstrate further contact's general ability to lessen prejudice" (p. 751). As [Hewstone \(2009\)](#) puts it, "[c]ontact works," and Allport's four conditions can be thought of as a "booster" to enhance efficacy (p. 247).

Five years ago, my colleagues and I published an updated meta-analysis on the contact hypothesis ([Paluck, Green, and Green 2019](#)). We exclusively analyzed randomized controlled trials with at least one day of delay between treatment and outcome

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measurement. Twenty-seven studies met these criteria. Among those studies, contact typically reduced prejudice. However, we also discovered four reasons for skepticism.

First, quality was inversely proportional to effectiveness: the larger the study, the smaller the effect, and studies with pre-analysis plans had a tiny, insignificant effect on average. Second, there was significant heterogeneity between targets of prejudice, with the most effective interventions aimed at people with disabilities. Third, we found troubling empirical gaps, including zero studies looking at interracial prejudice in people over 25. Fourth, there were no systematic variations of Allport's conditions in our dataset, and "a general lack of detailed description about the interventions" made it difficult to assess whether an intervention featured them at all ([Paluck, Green, and Green 2019](#), p. 141).

It has now been five years since that piece was published, and the question of what reduces prejudice has not lost any of its urgency. What have we learned since then?

I offer five observations on this theme. (For a recent systematic review, see [Clochard 2024](#).)

A welcome proliferation of field experiments (that find mixed effects)

Our meta-analysis coincided with the publication of impressive papers featuring rigorous and naturalistic contact interventions in high-tension settings. Two notable papers used intergroup sports to get groups to cooperate: cross-caste cricket in India ([Lowe 2021](#)) and interfaith soccer in Iraq ([Mousa 2020](#)). Another fruitful setting is schools, for instance exposing native-born Austrians to refugees for a classroom activity on fleeing one's home ([Nägele, Corcoran, and Klocke 2023](#)) or pairing "minority Roma and majority non-Roma Hungarians" students' seating assignments ([Elwert, Keller, and Kotsadam 2023](#)). Meanwhile, [Grady et al. \(2023\)](#) tested contact between farmers and pastoralists in middle belt Nigeria, a site of prior violent conflict. These studies comprise a new standard for credible contact research.

They also find disconcertingly mixed results. [Lowe \(2021\)](#) finds that playing together creates new friendships and reduces ingroup favoritism, while [Mousa \(2020\)](#) finds positive effects within the context of soccer, but very little that generalizes. [Nägele, Corcoran, and Klocke \(2023\)](#) find positive effects for contact, but also for the version of the intervention that lacked it. [Elwert, Keller, and Kotsadam \(2023\)](#) find strictly null results. [Grady et al. \(2023\)](#) find positive effects on stated attitudes, but no detectable effects on behaviors. Finally, [Scacco and Warren \(2018\)](#) studied computer training classes for Christians and Muslims in Nigeria, and found no effects on attitudes. The small effect on behavior they observe can be attributed to control subjects' becoming *more* discriminatory after experiencing within-group contact.

What is going on here? If contact typically reduces prejudice — if we can "answer the question of whether contact reduces prejudice with an emphatic 'yes'" ([Hewstone 2003](#), p. 352) — then what went wrong in these settings? Our meta-analysis inspired at least one critical rejoinder ([Pettigrew 2021](#), pp. 150-151), but it is *these* studies, and their middling findings, that most call for explanation.

Not all forms of prejudice are alike

In its common usage, prejudice connotes ‘pre-judgment:’ assessing someone before you know them based on group characteristics, or “statistical discrimination” ([Lenz and Mittlaender 2022](#)). In our meta-analysis, the most effective interventions were aimed at feelings that can plausibly be attributed to ignorance: people who are old ([Meshel and McGlynn 2004](#)) or blind ([Evans 1976](#)) or who have intellectual disabilities ([Di Tullio 1982](#)). The same cannot be said about [Scacco and Warren \(2018\)](#), where participants had experienced intergroup riots, or [Mousa \(2020\)](#), where Christians had been subject to ethnic cleansing at the hands of the Islamic State.

We need richer language for the many feelings that intergroup friction might engender. For instance, we might use **prejudice** for negative feelings that can plausibly be attributed to ignorance or a lack of experience; **hostility** for people who imagine a threat to existing social hierarchies or who have had negative but non-violent experiences with the outgroup; and **animus** for people who have experienced sustained discrimination and violence.

Add to this a bifurcated typology of contact, where **true acquaintance** meets Allport’s four conditions and **casual contact** does not. Allport would argue that *only* true acquaintance will move the needle, while [Pettigrew and Tropp \(2006\)](#) claim that the conditions are facilitating rather than essential.

I propose that casual contact can sometimes reduce prejudice if it dispels misconceptions, but is a poor fit for hostility, where it can be iatrogenic ([Enos 2014](#));³ that true acquaintance is required for reducing hostility; and that reducing animus is an unsolved problem. Unfortunately, the most intractable intergroup frictions — those motivated by bitter experience — are probably not amenable to contact. Rather, they recede only when a generation passes on or after an extraordinary period of sustained cooperation ([Jha 2013](#); [Jha and Shayo 2019](#)).

Ailment: type of dislike	Minimum viable treatment: type of contact
Prejudice: dislike based on ignorance (Evans 1976)	Casual contact: without Allport’s four conditions
Hostility: dislike based on negative experiences or threats to social hierarchy (Enos 2014)	True acquaintance: contact with shared goals, cooperation, equal status, and institutional support (Corno, La Ferrara, and Burns 2022)

³ [Hewstone \(2015\)](#) argues that [Enos \(2014\)](#) has “nothing whatsoever to do with intergroup contact” because it lacked “meaningful cross-group face-to-face interaction or opportunity to build a relationship and get beyond stereotypes” (p. 431). This same point, however, could be made about many cross-sectional studies that have been cited as evidence of contact’s long-term efficacy, where we have no specific knowledge about what (if any) interactions took place.

Animus: dislike based on violent negative experiences (Scacco and Warren 2018)	Unknown
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Table 1: a revised theory of prejudices and their appropriate contact treatment

Light touch interventions for light prejudice

Many contact scholars are pursuing what [Paluck et al. \(2021\)](#) call ‘light touch interventions’ defined by being quick, easy to implement, and cheap. For instance, we have a plethora of studies in which subjects imagine contact ([Crisp and Turner 2012](#)) or experience it vicariously ([Mazziotta, Mummendey, and Wright 2011](#)) and/or online ([Bond et al. 2023](#); [Stelzmann, Toth, and Schieferdecker 2021](#)).

These light touch interventions are well-suited for the lightest forms of prejudice: those not grounded in negative experiences. If we see evidence, for example, that contact with a video game character can change attitudes ([Mulak and Winiewski 2021](#)), it is fair to assume either that the subjects weren’t very hostile in the first place, or that outcomes do not capture attitudes of hostility or animus.

What role does assimilation play?

In the 1990s, queer theorists and activists had a lively debate about gay marriage. [Warner \(1999\)](#) writes that many “veterans of earlier forms of gay activism” would rank marriage as a much lower priority than “HIV and health care, AIDS prevention, the repeal of sodomy laws, antigay violence... and the saturation of everyday life with heterosexual privilege” (p. 120-121). Marriage, he writes, “has become the central legitimating institution by which the state penetrates the sexuality of its subjects” (p. 128).

In the end, the modern gay rights movement rallied around marriage, and thereby chose a squarely integrationist goal as its masthead. One lesson of this debate is that assimilation, at least the level of a movement or group, is partly a choice, and one that entails tradeoffs. The gay rights movement traded some amount of cultural distinctiveness for tolerance. I hypothesize that this experience generalizes, and assimilation is a key moderator of whether contact reduces prejudice.

[Broockman and Kalla \(2016\)](#) show that a brief conversation with a cisgendered or trans canvasser about trans rights can reduce anti-trans attitudes for at least 3 months. Does it matter how well the trans canvassers ‘pass’ as cis, i.e. how assimilated their gender performance is? How about whether they are advocating for a policy change that reinforces existing norms and social structures, like healthcare rights, versus something culturally disruptive, like competition in youth sports ([Goldbach et al. 2022](#); [Tanimoto and Miwa 2021](#))?

Likewise with antisemitism. The 14% of Jews who are strictly orthodox ([Staetsky 2022](#)) generally adhere to highly visible forms of non-assimilation. Do such signals affect the

degree of tolerance produced, and do effects generalize between different groups of Jewish people?

Towards systematic tests of the four conditions

[Elwert, Keller, and Kotsadam \(2023\)](#) seated non-Roma and Roma Hungarian students together in the classroom and found no meaningful changes in attitudes. However, being seated together does not guarantee that Allport's facilitating conditions were truly met; as the authors write, "equal status as students within the formal framework of Hungarian education may not prevent community prejudice from inflecting interactions among students, or between students and teachers" (p. 14).

This is a familiar problem undoubtedly because it's hard — the more naturalistic a field experiment, the less a researcher can hope to control individual interactions. Nevertheless, we would benefit from more designs that incorporate systematic variations of shared goals, cooperation, equal status, and institutional support (Clochard 2024). For example, [Greene et al. \(2023\)](#) induced "contact between participants with opposing partisan sympathies while experimentally varying their social status within the contact situation" (p. 2). The authors find that equal status contact leads to more tolerant behavior at a three week follow-up while unequal status contact does not.

Conclusion

Seventy years on, the contact hypothesis remains one of our strongest theories for how to reduce prejudice. The key lesson of contemporary contact research for policymakers, however, is that contact has a very mixed record against deep hostilities. To understand when and where contact will produce policy-relevant effects, I believe that Allport's four facilitating conditions remain essential, and that a richer theory of prejudice — one incorporating some aspects of the outgroup's choices, including willingness to assimilate — is called for.

This piece reflects solely the views of its author.

References

- Allport, G. W. (1954). *The nature of prejudice*.
- Bond, J., Dixon, J., Tredoux, C., & Andreouli, E. (2023). The contact hypothesis and the virtual revolution: Does face-to-face interaction remain central to improving intergroup relations? *Plos One*, 18(12), e0292831.
- Broockman, D., & Kalla, J. (2016). Durably reducing transphobia: A field experiment on door-to-door canvassing. *Science*, 352(6282), 220-224.
- Corno, L., La Ferrara, E., & Burns, J. (2022). Interaction, stereotypes, and performance: Evidence from South Africa. *American Economic Review*, 112(12), 3848-3875.

Clochard, G.-J. (2024). *Reducing prejudice: A Meta-Analysis and Experiment on the Contact Hypothesis*.

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Crisp, R. J., & Turner, R. N. (2012). The imagined contact hypothesis. *Advances in Experimental Social Psychology*, 46, 125-182.

Di Tullio, B. J. (1982). *The effect of employing trainable mentally retarded (TMR) students as workers within the Philadelphia public school system: Attitudes of supervisors and non-handicapped co-workers toward the retarded as a result of contact*. Temple University.

Elwert, F., Keller, T., & Kotsadam, A. (2023). Rearranging the desk chairs: A large randomized field experiment on the effects of close contact on interethnic relations. *American Journal of Sociology*, 128(6), 1809-1840.

Enos, R. D. (2014). Causal effect of intergroup contact on exclusionary attitudes. *Proceedings of the National Academy of Sciences*, 111(10), 3699-3704.

Evans, J. H. (1976). Changing attitudes toward disabled persons: An experimental study. *Rehabilitation Counseling Bulletin*.

Goldbach, C., Chambers-Baltz, S., Feeser, K., Milton, D. C., McDurmon, P., & Knutson, D. (2022). Transgender inclusion in competitive sport: Athletes' attitudes toward transgender athlete participation in intercollegiate sport. *International Journal of Sport and Exercise Psychology*, 1-23.

Grady, C., Wolfe, R., Dawop, D., & Inks, L. (2023). How contact can promote societal change amid conflict: An intergroup contact field experiment in Nigeria. *Proceedings of the National Academy of Sciences*, 120(43), e2304882120.

Greene, K., Rossiter, E., Seira, E., & Simpson, A. (2023). Interacting as equals: How contact can promote tolerance among opposing partisans. Retrieved May 17, 2024, from <https://ssrn.com/abstract=4456223>

Hewstone, M. (2003). Panacea for prejudice. *Psychologist*, 16(7), 352-355.

Hewstone, M. (2009). Living apart, living together? The role of intergroup contact in social integration. *Proceedings of the British Academy*, 162, 2008.

Hewstone, M. (2015). Consequences of diversity for social cohesion and prejudice: The missing dimension of intergroup contact. *Journal of Social Issues*, 71, 417-438. <https://doi.org/10.1111/josi.12120>

- Jha, S. (2013). Trade, institutions, and ethnic tolerance: Evidence from South Asia. *American Political Science Review*, 107(4), 806-832.
- Jha, S., & Shayo, M. (2019). Valuing peace: The effects of financial market exposure on votes and political attitudes. *Econometrica*, 87(5), 1561-1588.
- Lenz, L., & Mittlaender, S. (2022). The effect of intergroup contact on discrimination. *Journal of Economic Psychology*, 89, 102483.
- Lowe, M. (2021). Types of contact: A field experiment on collaborative and adversarial caste integration. *American Economic Review*, 111(6), 1807-1844.
- Mazziotta, A., Mummendey, A., & Wright, S. C. (2011). Vicarious intergroup contact effects: Applying social-cognitive theory to intergroup contact research. *Group Processes & Intergroup Relations*, 14(2), 255-274.
- Meshel, D. S., & McGlynn, R. P. (2004). Intergenerational contact, attitudes, and stereotypes of adolescents and older people. *Educational Gerontology*, 30(6), 457-479.
- Mousa, S. (2020). Building social cohesion between Christians and Muslims through soccer in post-ISIS Iraq. *Science*, 369(6505), 866-870.
- Mulak, A., & Winiewski, M. H. (2021). Virtual contact hypothesis: Preliminary evidence for intergroup contact hypothesis in interactions with characters in video games. *Cyberpsychology: Journal of Psychosocial Research on Cyberspace*, 15(4), Article 6. <https://doi.org/10.5817/CP2021-4-6>
- Nägele, S., Corcoran, K., & Klocke, U. (2023). It is worth taking a closer look: A field experiment on intergroup contact between Austrian pupils and refugees. *Group Processes & Intergroup Relations*, 13684302231180828.
- Paluck, E. L., Green, S. A., & Green, D. P. (2019). The contact hypothesis re-evaluated. *Behavioural Public Policy*, 3(2), 129-158.
- Paluck, E. L., Porat, R., Clark, C. S., & Green, D. P. (2021). Prejudice reduction: Progress and challenges. *Annual Review of Psychology*, 72, 533-560.
- Pettigrew, T. F. (2021). Applying contextual analyses to intergroup contact. In T. F. Pettigrew, *Contextual social psychology: Reanalyzing prejudice, voting, and intergroup contact* (pp. 121-156). American Psychological Association. <https://doi.org/10.1037/0000210-008>
- Pettigrew, T. F., & Tropp, L. R. (2006). A meta-analytic test of intergroup contact theory. *Journal of Personality and Social Psychology*, 90(5), 751.

Scacco, A., & Warren, S. S. (2018). Can social contact reduce prejudice and discrimination? Evidence from a field experiment in Nigeria. *American Political Science Review*, 112(3), 654-677.

Staetsky, L. D. (2022). Haredi Jews around the world: Population trends and estimates. *Institute for Jewish Policy Research: London, UK*.

Stelzmann, D., Toth, R., & Schieferdecker, D. (2021). Can intergroup contact in virtual reality (VR) reduce stigmatization against people with schizophrenia? *Journal of Clinical Medicine*, 10(13), 2961.

Tanimoto, C., & Miwa, K. (2021). Factors influencing acceptance of transgender athletes. *Sport Management Review*, 24(3), 452-474.

Warner, M. (1999). Normal and normaller: Beyond gay marriage. *GLQ: A Journal of Lesbian and Gay Studies*, 5(2), 119-171.